

GISBORNE, New Zealand

WMO: 932920

Lat: 38.6586S

Lon: 177.9850E

Elev: 5

StdP: 101.26

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.3	3.2	-0.2	3.7	5.1	0.8	4.0	5.5	10.0	13.6	9.0	12.9	2.1	330	0.454

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.1	27.9	19.5	26.5	18.9	25.2	18.4	20.9	25.4	20.1	24.3	19.4	23.4	5.3	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.5	14.2	22.5	18.6	13.5	21.8	17.8	12.8	21.4	60.5	25.5	57.7	24.4	55.2	23.5	24.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.3	8.1	7.2	DB	0.5	31.4	0.9	2.0	-0.1	32.8	-0.7	33.9	-1.1	35.0	-1.8	36.5
			WB	0.0	22.5	1.0	0.9	-0.7	23.1	-1.2	23.6	-1.8	24.1	-2.4	24.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.4	19.2	19.0	17.5	15.1	12.9	10.5	9.9	10.3	12.0	13.8	15.5	17.9	(6)
(7)		DBStd	4.15	2.86	2.66	2.41	2.55	2.58	2.41	2.25	2.26	2.47	2.57	2.83	2.71	(7)
(8)		HDD10.0	90	0	0	0	1	4	22	29	23	8	2	0	0	(8)
(9)		HDD18.3	1574	23	21	46	102	170	236	261	248	191	142	93	42	(9)
(10)		CDD10.0	1710	287	253	231	153	93	36	27	34	68	119	164	244	(10)
(11)		CDD18.3	153	52	41	19	4	1	0	0	0	0	1	7	28	(11)
(12)		CDH23.3	802	316	225	86	10	1	0	0	0	0	4	33	127	(12)
(13)		CDH26.7	113	56	37	6	0	0	0	0	0	0	1	2	12	(13)
(14)	Wind	WSAvg	3.1	3.3	2.9	2.9	2.6	2.8	2.9	3.0	3.3	3.6	3.7	3.3	(14)	
(15)	Precipitation	PrecAvg	1032	56	71	92	97	89	128	114	107	91	59	59	72	(15)
(16)		PrecMax	1388	146	189	322	239	290	328	326	294	238	115	122	177	(16)
(17)		PrecMin	739	10	4	8	13	19	40	22	24	11	15	14	5	(17)
(18)		PrecStd	174	35	47	66	60	60	65	64	66	57	26	28	46	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.4	30.2	27.2	24.6	22.1	19.5	18.0	18.7	21.6	23.5	26.2	28.2	(19)
(20)			MCWB	20.0	20.8	18.7	18.5	16.9	15.1	13.7	13.7	15.3	15.8	17.7	19.3	(20)
(21)		2%	DB	28.1	27.6	25.5	22.7	20.3	17.6	16.6	17.2	19.7	21.7	24.1	26.1	(21)
(22)			MCWB	19.5	19.9	18.3	17.0	15.6	13.6	12.3	12.4	14.2	15.3	16.9	18.4	(22)
(23)		5%	DB	26.6	26.0	24.1	21.2	19.0	16.4	15.4	16.0	18.2	20.3	22.5	24.7	(23)
(24)			MCWB	18.9	19.3	17.8	16.3	14.9	12.8	11.7	11.7	13.2	14.5	16.3	18.0	(24)
(25)		10%	DB	25.0	24.4	22.6	19.9	17.8	15.3	14.3	14.9	17.0	19.0	21.1	23.3	(25)
(26)			MCWB	18.3	18.8	17.5	16.0	14.1	12.0	11.1	11.2	12.5	13.9	15.4	17.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.9	22.6	20.7	19.6	18.0	16.1	14.6	14.7	16.0	17.1	19.0	20.6	(27)
(28)			MCDWB	27.5	26.7	24.5	22.5	20.2	18.3	17.0	17.4	20.3	22.0	24.4	25.8	(28)
(29)		2%	WB	20.8	21.2	19.5	18.4	16.8	14.6	13.2	13.3	14.8	15.9	17.7	19.5	(29)
(30)			MCDWB	25.7	25.3	22.8	20.9	19.3	16.6	15.4	15.7	18.3	20.4	22.5	24.3	(30)
(31)		5%	WB	19.9	20.3	18.7	17.4	15.6	13.5	12.4	12.4	14.0	15.1	16.9	18.7	(31)
(32)			MCDWB	24.4	24.3	22.2	19.9	18.1	15.6	14.5	14.8	17.3	19.2	21.4	23.2	(32)
(33)		10%	WB	19.1	19.5	18.0	16.6	14.6	12.4	11.6	11.6	13.1	14.4	16.0	17.9	(33)
(34)			MCDWB	23.4	23.3	21.6	19.1	17.1	14.8	13.8	14.2	16.2	18.2	20.4	22.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.1	8.7	8.7	8.7	8.5	8.2	7.6	8.3	8.7	8.9	8.8	8.6	(35)
(36)			MCDBR	12.1	11.4	11.5	10.6	9.8	9.4	8.9	9.8	10.6	10.7	11.2	11.0	(36)
(37)			MCWBR	5.4	5.4	5.8	6.0	6.0	6.2	5.8	6.1	6.4	6.1	6.1	5.6	(37)
(38)		5% WB	MCDBR	10.0	9.3	9.2	8.0	7.8	7.7	7.2	8.3	9.3	9.5	10.1	9.4	(38)
(39)			MCWBR	5.0	5.0	5.3	5.0	5.3	5.7	5.2	5.7	6.0	5.8	5.9	5.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.348	0.343	0.330	0.314	0.301	0.295	0.298	0.300	0.322	0.334	0.335	0.347	(40)	
(41)		taud	2.539	2.575	2.606	2.639	2.647	2.634	2.627	2.627	2.537	2.514	2.531	2.538	(41)	
(42)		Ebn at Noon	984	965	938	892	845	818	838	895	929	963	991	989	(42)	
(43)		Edh at Noon	108	100	89	75	65	60	65	75	94	105	108	109	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.67	5.62	4.46	3.19	2.27	1.78	1.94	2.80	3.94	5.29	6.11	6.47	(44)	
(45)		RadStd	0.41	0.40	0.37	0.23	0.17	0.12	0.15	0.18	0.38	0.30	0.40	0.46	(45)	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46) Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (4 neighbors)		+0.25	+0.35	+0.43	N/A	N/A	+0.33	-18	-76	+86	N/A
		+0.29	+0.38	+0.57	+0.46	N/A	N/A	-17	-84	+86	N/A

Nomenclature: See separate page